BOJJI ANIRUDH

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SUMMARY

Software Engineer with 2+ years of experience building test automation and validation frameworks for multi-cloud networking infrastructure across AWS, GCP, Azure, and OCI. Hands-on with dataplane and routing systems (VPP, DPDK, BGP, IPSec, VXLAN, FRR), CI/CD with Jenkins on Kubernetes, and applied AI/ML for operational tooling including log analysis, anomaly detection, and AI-assisted DevOps workflows. Co-author of four peer-reviewed publications in graph machine learning and NLP.

SKILLS

Cloud Platforms: AWS, GCP, Azure, OCI

Networking: BGP, IPSec, VXLAN, VPP, DPDK, FRR, Multi-cloud Networking, REST APIs

Languages: Python, C, Go, Bash, Java, SQL

Infrastructure & DevOps: Terraform, Kubernetes, Jenkins, Git, Linux, Docker

Automation & Testing: Robot Framework, Python Test Automation, MCP Tools, Claude Code

AI / ML: TensorFlow, CNNs, Graph Neural Networks, LLM-based Log Analysis, Anomaly Detection

Databases: MySQL, PostgreSQL

EXPERIENCE

Software Engineer — Alkira Networks | Jun 2025 – Present

Bengaluru, Karnataka, India

- Own test automation for dataplane and routing systems (VPP/DPDK, BGP, IPSec, VXLAN) across a multi-cloud networking platform serving enterprise customers.
- Built MCP tools and Claude Code skill files for internal DevOps workflows, enabling AI-assisted network debugging, health checks, and tenant diagnostics — reducing manual toil for engineering teams.
- Design and execute end-to-end validation for cloud connectors across AWS, GCP, Azure, and OCI.
- Build internal test tooling in Python, Go, and Bash for node validation and network debugging.
- Drive release quality through regression automation, performance benchmarking, and production readiness testing.
- Validate service insertion workflows for inline security services (Palo Alto Networks, Cisco FTDv, Zscaler), ensuring correct traffic steering, policy enforcement, and failover behavior. Linked PAN, Cisco FTDv, and Zscaler service instances with network segments, verifying end-to-end traffic flow through firewalls and secure web gateways across multi-cloud environments.
- Test and verify scaling operations (scale-out/scale-in) for network nodes, validating tunnel re-convergence, BGP session re-establishment, traffic policy redistribution, and symmetric data-plane state across scaled nodes.

Associate Software Engineer — Alkira Networks | Jul 2024 – Jun 2025

Bengaluru, Karnataka, India

- Triaged and resolved customer-found defects (CFDs), reproducing issues in multi-cloud environments, root-causing failures across dataplane and routing layers, and driving fixes to production.
- Built automated bandwidth and performance testing across AWS, GCP, and Azure, validating multi-cloud network throughput and surfacing regressions before customer impact.
- Designed and deployed an AI-powered log analyzer that summarizes errors, detects anomalies, and accelerates incident triage for the on-call team.
- Developed end-to-end test automation suites using Python and Robot Framework, improving regression coverage and shortening release validation cycles.

- Engineered the alerting flow for production cloud-networking services, tuning signal-to-noise and integrating alerts with on-call workflows.
- Set up Jenkins-on-Kubernetes CI/CD system, enabling scalable containerized test pipelines across engineering teams.
- Validated backend REST APIs and networking features involving BGP and IPSec connectivity.

Amazon ML Summer School — Amazon | Sep 2023 – Oct 2023

- Selected for Amazon's ML Summer School 2023, gaining hands-on experience with supervised and unsupervised learning, deep learning, NLP, and computer vision techniques using real-world datasets and Amazon's ML stack.

Cyber Security Intern — Cyber Ekta | Apr 2023 – Jul 2023

Hyderabad, India

- Performed network and application security assessments using Nmap, Nessus, Wireshark, and Burp Suite, identifying and reporting vulnerabilities.

PROJECTS

Terabit-Scale Network-on-Chip Router (SystemVerilog)

Designed a 2D-mesh NoC router using deflection (hot-potato) routing, XY dimension-ordered routing, and credit-based flow control. Verified cycle-accurately with Verilator testbenches — 700K+ flits simulated with zero loss — and characterized latency vs. load through saturation. github.com/Anirudhbojji/noc-deflection-router

Real-Time BGP Hijack Detector

Streams the global routing table via RIPE RIS Live, validates announcements against RPKI, and alerts on origin and sub-prefix hijacks of watched prefixes with webhook integration. github.com/Anirudhbojji/bgp-hijack-detector

PCAP Threat Hunter (Mini-IDS)

Detects port scans, C2 beaconing (inter-arrival variance analysis), DNS tunneling (entropy scoring), and cleartext credentials in packet captures; validated end-to-end against generated attack traffic. github.com/Anirudhbojji/pcap-threat-hunter

Chaos Engineering for Network Resilience

Built a chaos testing framework that injects controlled failures — tunnel flaps, BGP peer drops, and node crashes — into multi-cloud network environments. Validates automatic failover, measures convergence times, and verifies traffic rerouting across redundant paths.

AWS Cloud Compliance & Security Auditor

Built a Python tool that runs hourly across multiple AWS accounts to audit EC2, RDS, and S3 resources for missing or non-compliant tags, open security groups, overly permissive IAM policies, and publicly exposed resources. Identifies violations, maps them to owners, and sends compliance reports via webhook notifications.

Automated Multi-Cloud Connectivity Lab

Built a self-service lab environment that spins up end-to-end network connectivity across AWS and GCP using Terraform. Automated provisioning of VPCs, subnets, EC2/GCE instances, IPSec site-to-site tunnels, and BGP route exchange between cloud regions. Includes traffic validation scripts to verify reachability and measure throughput.

GitHub User Classification Using Graph Neural Networks

Built a Graph Convolutional Network classifying GitHub users (ML vs. web developers) on a 37K-node social graph; GCN embeddings with gradient boosting reached 85.5% accuracy. Published in the Indian Journal of Technical Education (ISTE), Vol. 47, No. 2, 2024. github.com/Anirudhbojji/github-user-classification-gcn

PUBLICATIONS

- “Utilizing Graph Convolutional Networks for GitHub User Classification” — Indian Journal of Technical Education (ISTE), Vol. 47, No. 2, 2024
- “Enhancing Telugu Sarcasm Classification Models with Word Embeddings in Imbalanced Datasets” — FICTA 2024, Springer

- “Enhancing Telugu Abusive Language Detection Using Word Embeddings and BERT Models” — 2nd Int’l Conf. on Recent Trends in Microelectronics, 2024
- “Automated Detection of Offensive Content in Social Media” — IRSD 2024
- Google Scholar: scholar.google.com/citations?user=QZ075QoAAAAJ

EDUCATION

B.E. Computer Science Engineering — Osmania University | 2020 – 2024

CGPA: 8.1 / 10

CERTIFICATIONS

- Alkira Network Infrastructure as-a-Service (Expert Level) — Alkira, Inc. (Jan 2025)
- Google Associate Cloud Engineer — Google (May 2025)
- Microsoft Certified: Azure Fundamentals — Microsoft (Apr 2023)